

Manuel Buenrostro Macedo

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EDUCATION

Embry-Riddle Aeronautical University

B.S. Aerospace Engineering, Rocket Propulsion | GPA: 3.102 / 4.0

Expected May 2028

Daytona Beach, FL

EXPERIENCE

Student Worker, Low Gravity System Design

Jan. 2026 – Present

Robotics and Intelligent Learning Lab (RILL), Embry-Riddle Aeronautical University

Daytona Beach, FL

- Co-authored and submitted a conference paper to IEEE and IROS on a 10-DoF 3D-printed lower-limb exoskeleton, writing major sections on cycloidal actuator design, testing, and validation.
- Designed 18 custom parts and built actuator/exoskeleton test infrastructure in Autodesk Inventor, including a dynamometer-based torque rig and suspended/treadmill-assisted test setup.
- Printed, assembled, repaired, and tested 3 cycloidal actuators with 13:1 reduction using PETG-CF, PLA-CF, nylon-CF, aluminum, and custom electrical assemblies.
- Conducted 40–50 test runs using ODrive S1, ROS2, Python, CAN bus, Raspberry Pi 4B, KB2040, and a DYN-210 torque sensor to validate onboard torque feedback against external measurements.
- Performed electrical integration and repair of exoskeleton subsystems, including CAN wiring, power wiring, emergency stops, soldered connections, and electrical bus repairs to support safe human-subject testing.

Project Lead / Principal Investigator; Undergraduate Research Assistant

Sep. 2024 – Present

Engineering Physics Propulsion Lab (EPPL), Embry-Riddle Aeronautical University

Daytona Beach, FL

- Lead OpenArm project planning, prototyping, and technical presentation within EPPL, supporting modular manipulator development for quadruped robotics research.
- Supported iterative subsystem design, fabrication, troubleshooting, and live demonstrations across robotics research platforms, communicating technical progress to faculty, visitors, and sponsors.
- Integrated and tested research hardware while refining mechanical parts in Autodesk Inventor for lab research platforms and showcase events.

PROJECTS

OpenArm — Modular Robotic Manipulator | Project Lead, EPPL

Jan. 2025 – Nov. 2025

- Led CAD design of OpenArm for the OpenMutt quadruped platform and developed OpenHand, a compliant 5-bar linkage gripper with a continuous-rotation wrist.
- Built 2 arm prototypes and 2 subsystem testbeds using Autodesk Inventor, Arduino IDE, Visual Studio Code, FDM/resin printing, soldering, and electrical diagnostic tools.
- Designed and demonstrated a quick-disconnect end-effector interface; integrated the manipulator on OpenMutt with 4 DoF and an approximate 1 m reachable workspace.
- Presented OpenArm at the Fall Student Research Symposium through a poster and live demonstration, showcasing modular manipulation capability for quadruped robotics research.

OpenMutt — Open-Source Quadruped

Sep. 2024 – Present

- Developed a pan-tilt multi-sensor mount and quick-disconnect end-effector mounting system using CAD and additive manufacturing.

VEX Robotics Competition — Team 7121E

Mar. 2020 – May 2024

- Led a 7-member team to 21 regional, 3 state, and 1 world championship appearances, earning 7 design/merit awards.

SKILLS

Software & Programming: Autodesk Inventor, CATIA V5, Python, MATLAB, ROS2, Arduino IDE, Visual Studio Code, ODrive GUI

Hardware & Systems: CAN bus, Raspberry Pi 4B, ODrive S1, KB2040 microcontrollers, electrical integration, soldering, multimeter diagnostics

Manufacturing & Testing: FDM/resin 3D printing, additive manufacturing, actuator testing, dynamometer testing, data logging, rapid prototyping

Technical Communication: Technical writing, grant writing, engineering documentation, technical presentations

Languages: English, Spanish

HONORS & PUBLICATIONS

SURF Grant Recipient: Embry-Riddle Summer Undergraduate Research Fellowship, 2026

Co-author: Submitted conference paper to IEEE and IROS on a low-cost 3D-printed full lower-limb exoskeleton, 2026

Presenter: Fall Student Research Symposium, OpenArm poster and live demonstration, Nov. 2025

Scholarships: Presidential Scholarship, Embry-Riddle Aeronautical University; Florida Bright Futures Scholarship, Full Award